

CONCORDIA UNIVERSITY
Department of Computer Science
SOEN 341/4 S Software Process

MID TERM EXAM: February 19, 2016

8:45 -10:00 a.m. – (1 hour 15 min)

Question 1 System Sequence Diagram (10 marks)

Draw the system sequence diagram of the following case in a student course planning system: The precondition is that the student is logged into the system and is presented with a home page.

CASE: A student wishes to add a course

Your system sequence diagram should show the interaction of the system with the student for the selection of a course in a given semester. You must give a dialog illustrating the detail of selection, for example: semester, course, section, result, etc.

Question 2 Full Sequence Diagram (30 marks)

Referring to the diagram of question 1, draw a full sequence diagram of the above interaction, using the objects you consider appropriate to handle all aspects of the selection operation. Label both the sent message lines and the return lines.

Note that the system must have all information to ensure that there is no time conflict, the prerequisites are met and that there is space in the class.

Question 3 Class diagram (30 marks)

Referring to your sequence diagram of question 2, draw a class diagram showing the classes needed to model the system. Show the associations between the classes with appropriate labels, Model the classes with three sections: name, attributes and methods. Add some attributes and methods to some classes, but do not try to make fully complete entries.

Give a brief description of the diagram in a few lines of text.

Question 4 Process (30 marks)

Concordia University has given you a contract of 20 million dollars to develop a new Information System (SIS). The new information system aims to combine many old information systems to serve as one system that provides all features from the old systems. The new system must support access from web, mobile web, mobile app (iOS and Android) and desktop app (Windows and MacOS). The university allows no service interruption during the period of the new system development and wants to avoid any drastic changes to the system at any time to keep the daily work of the university as usual.

Your company has 100 employees including 40 lively and talented programmer analysts as well as others specialised in various aspects of management, quality control, and

documentation. Around half of the employee have experience in the design and development of some of the old systems.

In no more than a page, state: 1) the process of that you would follow for the project and 2) the architecture of the new system. Explain the rationale of your choices.